OpenSource Compass

Product Requirements Document (PRD)

Version: 1.0

Status: Draft

Document Owner: Nihaal N O

Last Updated: June 2026

Table of Contents

1. Executive Summary
2. Vision
3. Problem Statement
4. Product Goals
5. Target Audience
6. User Personas
7. User Journey
8. Functional Requirements
9. Non-Functional Requirements
10. AI Capabilities
11. Dashboard Modules
12. Technical Stack
13. Success Metrics
14. Future Scope
15. Out of Scope (Version 1)

---

1. Executive Summary

OpenSource Compass is an AI-powered developer platform designed to simplify the process of contributing to open-source software.

The platform analyzes a developer's GitHub profile, understands their technical skills, recommends suitable repositories, identifies beginner-friendly issues, and provides AI-assisted explanations and contribution guidance.

The objective is to significantly reduce the learning curve associated with open-source contribution while helping developers build stronger GitHub portfolios.

---

2. Vision

To become the most intelligent developer platform for discovering, understanding, and contributing to open-source software.

Rather than acting as another repository search engine, OpenSource Compass aims to function as an AI mentor that helps developers successfully complete meaningful contributions.

---

3. Problem Statement

Many developers encounter significant barriers when attempting to contribute to open source.

Common challenges include:

- Finding repositories appropriate for their skill level
- Understanding unfamiliar codebases
- Choosing issues that match their experience
- Navigating large project structures
- Learning contribution workflows
- Building a professional GitHub profile

Existing solutions expose repository data but provide little guidance on where developers should begin or how they should approach contributions.

---

4. Product Goals

Primary Goals

- Simplify open-source onboarding.
- Reduce repository learning time.
- Personalize repository discovery.
- Recommend suitable contribution opportunities.
- Increase successful pull requests.

Secondary Goals

- Encourage consistent contributions.

- Improve developer confidence.
- Track contribution growth.
- Build long-term developer portfolios.

---

5. Target Audience

Primary Users

- College students
- Beginner developers
- Self-taught programmers
- First-time open-source contributors

Secondary Users

- Professional developers
- Technical mentors
- Open-source maintainers
- Coding bootcamp participants

---

6. User Personas

Persona A — Student

Objective

Contribute to GitHub projects but lacks practical experience.

Pain Points

- Doesn't know where to start.
- Finds repositories overwhelming.
- Cannot identify beginner-friendly issues.

Persona B — Intermediate Developer

Objective

Find projects aligned with current technical skills.

Pain Points

- Too many repositories.
- Unsure which projects provide meaningful experience.
- Wants contributions that improve career prospects.

Persona C — Maintainer

Objective

Receive higher-quality pull requests.

Pain Points

- Contributors lack project understanding.
- Frequent duplicate or incomplete pull requests.
- High onboarding effort.

---

7. User Journey

Landing Page

↓

GitHub Authentication

↓

Profile Analysis

↓

Skill Extraction

↓

Repository Recommendations

↓

Repository Exploration

↓

Issue Recommendation

↓

AI Repository Explanation

↓

Contribution Planner

↓

Pull Request Submission

↓

Contribution Dashboard

---

8. Functional Requirements

8.1 Authentication

Features

- GitHub OAuth
- User profile
- Session management
- Secure authentication

8.2 GitHub Profile Analysis

Collect:

- Languages
- Frameworks
- Public repositories
- Commit activity
- Stars
- Forks
- Contribution graph

- Repository topics

8.3 Skill Analysis Engine

Generate:

- Primary programming languages
- Framework preferences
- Experience estimation
- Skill score
- Technology recommendations

8.4 Repository Recommendation Engine

Recommend repositories based on:

- Programming language
- Technology stack
- Skill match
- Difficulty
- Repository activity
- Community size
- Repository health

8.5 Repository Explorer

Display:

- Repository overview
- README summary
- Technology stack
- Stars
- Forks
- Contributors
- License
- Releases
- Branches

8.6 AI Repository Understanding

Generate:

- Plain-language repository summary
- Architecture overview
- Folder explanations
- Entry point identification
- Important files
- Dependency relationships

8.7 Issue Recommendation Engine

Recommend issues using:

- Skill compatibility
- Difficulty level
- Labels
- Good First Issue
- Help Wanted
- Estimated effort
- Activity level

8.8 AI Issue Explanation

Generate:

- Simplified explanation
- Required knowledge
- Relevant files
- Suggested implementation approach
- Expected learning outcome
- Estimated completion time

8.9 Personalized Learning Roadmap

Provide:

- Current skill assessment
- Missing technologies
- Learning sequence
- Suggested repositories

- Recommended contribution milestones

8.10 Contribution Dashboard

Track:

- Pull requests opened
- Pull requests merged
- Issues solved
- Repository contributions
- Programming languages
- Contribution streak
- Growth statistics

8.11 Notifications

Notify users about:

- New repository recommendations
- Matching issues
- Pull request updates
- Maintainer responses
- Trending repositories

---

9. Non-Functional Requirements

Performance

- Landing page under 2 seconds
- Repository analysis under 15 seconds
- AI responses under 10 seconds

Security

- OAuth authentication
- Secure token management
- HTTPS
- Encrypted credentials

Scalability

- Support 100,000+ users
- Horizontal backend scaling
- Repository caching

Availability

Target uptime:

99.9%

---

10. AI Capabilities

The AI system will assist users by providing:

- Repository summaries
- Architecture explanations
- Folder explanations
- Issue explanations
- Learning recommendations
- Contribution planning
- Beginner guidance

---

11. Dashboard Modules

The dashboard will contain:

- Home
- Repository Recommendations
- Repository Explorer
- Recommended Issues
- Learning Roadmap
- Contribution Dashboard
- Notifications
- Profile
- Settings

---

12. Technical Stack

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Framer Motion

Backend

- Node.js
- Express.js

Database

- PostgreSQL
- Supabase

Authentication

- Supabase Authentication
- GitHub OAuth

APIs

- GitHub REST API
- GitHub GraphQL API

Artificial Intelligence

- OpenAI
- Gemini
- Ollama (optional local inference)
- Retrieval-Augmented Generation (RAG)

Deployment

- Vercel
- Docker

---

13. Success Metrics

The following KPIs will measure product success:

- Daily active users
- Weekly active users
- Repository recommendation click-through rate
- Pull requests created
- Pull requests merged
- Repository onboarding time
- AI explanation satisfaction
- User retention

---

14. Future Scope

The following features are planned for future versions:

- VS Code Extension
- Browser Extension
- AI Pull Request Review
- AI Codebase Visualization
- Team Workspaces
- Organization Dashboards
- Maintainer Portal
- Contribution Certificates
- Hackathon Mode
- Hacktoberfest Integration
- Community Discussions
- Mobile Application

---

15. Out of Scope (Version 1)

The following capabilities are intentionally excluded from Version 1:

- Git repository hosting
- Integrated IDE
- CI/CD pipeline management
- Self-hosted Git providers
- Live collaborative coding
- Project management features
- Team chat

Product Vision Statement

OpenSource Compass aims to bridge the gap between developers and open-source communities by combining GitHub data, intelligent recommendation systems, and AI-powered guidance into a single developer-first platform.

Version 1 focuses on delivering an intuitive, production-ready experience that helps users discover suitable repositories, understand unfamiliar codebases, and confidently make meaningful open-source contributions.